Lab1 Questions and Solutions

[Platform → Hackerrank]

Q1 .[Easy Squares]A number 'n' is called a perfect square if it can be expressed as a product of 2 equal integers, let say x. In other words, the equation: x * x = n must hold for some integer x and n. For example, if n = 25, then x = 5 will satisfy the equation. For a given perfect square number n, find integer x.

Input :

The first line contains an integer denoting perfect square n.

Output:

For the first line, print an integer x satisfying the equation x * x = n

Sample Input 1 :

25

Sample Output 1:

5

Sample Input 2 :

625

Sample Output 2 :

25

Solution

```
n = int(input())
x = n**0.5
print(int(x))
```

Q2.[Help Me Newton] A constant retarding force of X Newton is applied to a body of mass Y kg moving initially with a speed of Z m/s. Print how long does the body take to stop?

Formulae to be used:

As the retarding force X is constant, the acceleration will also be constant throughout(Why? - as mass Y is constant, the acceleration, a = X/Y will also be constant)

Since the body comes to complete halt after some time t, its final velocity, V can be assigned as 0.

Therefore, the total time as required in the question, can be calculated as:

t= Z/a [From first equation of motion, $v = u + at$]

Input :

The first line contains the value of x

The second line contains the value of y
The third line contains the value of z

Output:
Print how long does the body take to stop

Sample Input 1 :
50
20
15
Sample Output 1:
6
Explanation:
v = 0;
a = F/m = x/y = -50/20 = -2.5
t = (v-u)/a = (0-z)/(-2.5) = -15/(-2.5) = 6

Solution
```
X=int(input())
Y=int(input())
Z=int(input())
a = X/Y
t = Z/a
print(t)
```

Q3. [Discriminating Quadratic]The standard form of quadratic equation is defined as : ax^2 + bx + c. For a given quadratic equation with given a, b and c, find its discriminant. (Discriminant is defined as b^2 - 4ac).
Input :
The first 3 lines contain integers denoting a,b and c.
Output:
Print the discriminant of the polynomial

Sample Input 1 :
1
-10
25
Sample Output 1:
0

Solution:

```
a = int(input())
b = int(input())
c = int(input())
D = (b*b) - (4*a*c)
print(D)
```

Q4:[That's My Interest] A simple interest on a Principal amount can be calculated as $(P * R * T)/100$, where:

- P- **Principal amount**
- R - **Rate of Interest**
- T - **Time period.**

The user takes input for P, R, and T respectively, and compute and prints the Simple Interest- S.

Input :

The first 3 lines contain integers denoting P, R, and T

Output:

Print the Simple Interest- S corresponding to P, R, and T

Sample Input 1 :

12

25

4

Sample Output 1:

12

Solution:

```
P = int(input())
R = int(input())
T = int(input())
S = (P*R*T)/100
print(S)
```